

Project Purpose, Real-World Usage, And Value

What The Project Does

The project is an Intelligent Smart Microgrid Energy Management System. It combines weather data, realistic demand simulation, battery safety logic, machine learning forecasting, reinforcement learning dispatch, backend APIs, and an operator dashboard.

At a high level, the system answers one operational question:

Given expected solar generation, expected load, current battery SoC, and tariff, should the microgrid charge the battery, discharge the battery, idle, import from grid, or raise an alert?

The implemented system has these major layers:

Layer	What it does
Data engineering	Loads NASA POWER weather data, cleans it, standardizes columns, interpolates missing values, sorts timestamps, removes duplicates, and normalizes features.
Synthetic operations layer	Adds realistic load demand, India-style time-of-use tariff, battery SoC, grid import, and derived temporal features.
Forecasting layer	Trains LSTM models on GPU to forecast solar generation and load demand.
Optimization layer	Trains a PPO reinforcement learning agent in a custom microgrid environment.
Dispatch engine	Converts solar/load/tariff/SoC conditions into operator-readable charge, discharge, or idle decisions.
Alerts layer	Detects peak demand risk, renewable drops, battery limit risk, and peak tariff import exposure.
Metrics layer	Computes cost savings, renewable utilization, grid dependency, self-sufficiency, and battery health.
API layer	Exposes FastAPI endpoints for forecast, optimization, decisions, metrics, alerts, scenarios, MATLAB export, and live simulation.
Dashboard layer	Gives grid operators a control-room view with live KPIs, charts, recommendations, alerts, scenario simulation, and override controls.

Real-World Scenario

A practical deployment target is a campus, industrial facility, hospital, data center, commercial building, or rural microgrid with:

Asset	Example
Solar PV	Rooftop or ground-mounted solar generation.
Battery	Lithium-ion BESS connected through inverter/PCS.

Asset	Example
Grid connection	Import from utility feeder under time-of-use tariff.
Critical load	Facility demand that must be served reliably.
Operator	Energy manager or control-room staff monitoring dispatch decisions.

In a real microgrid, solar generation changes with cloud cover, temperature, humidity, and time of day. Load changes with occupancy, HVAC, working hours, weekday/weekend behavior, and events. Tariff changes by time block. Battery operation must respect physical limits. The EMS coordinates these factors so the battery is not used blindly.

Real Operational Use Cases

1. Peak Tariff Reduction

Evening tariff is expensive while solar is low. The EMS can preserve or pre-charge the battery before evening and discharge during the peak tariff block.

Operational value:

- Reduces grid purchase during expensive periods.
- Prevents battery depletion before the most valuable dispatch window.
- Gives the operator a reason string instead of a black-box command.

2. Renewable Self-Consumption

When solar is higher than load, the EMS charges the battery instead of curtailing useful renewable energy.

Operational value:

- Increases renewable utilization.
- Reduces wasted PV generation.
- Improves sustainability metrics.

3. Peak Demand Risk Detection

The alert engine identifies forecasted or current high-load windows.

Operational value:

- Gives the operator time to prepare.
- Supports demand response decisions.
- Prevents avoidable high grid import.

4. Battery Safety Monitoring

The system enforces battery SoC limits. Current implementation uses:

Parameter	Value
Battery capacity	180 kWh

Parameter	Value
Minimum SoC	20 percent
Maximum SoC	90 percent
Max charge power	55 kW
Max discharge power	58 kW
Round-trip efficiency	93 percent

Operational value:

- Prevents over-discharge and over-charge in simulation.
- Tracks equivalent cycles.
- Produces a battery health score.

5. Operator Training Without Live SCADA

Because real meter/BMS/SCADA access is not available, the project includes a backend live simulator. The dashboard can switch from static historical data to dynamic live telemetry at a 1.5 second refresh cadence.

Operational value:

- Demonstrates a control-room workflow without hardware.
- Allows scenario testing.
- Gives reviewers a realistic live EMS experience.

How It Is Better Than Current Basic Solutions

The project should not be presented as a replacement for certified industrial SCADA. It is better framed as an intelligent EMS prototype that improves over basic rule-based or spreadsheet-style dispatch.

Current/basic solution	Limitation	Project improvement
Manual operator decisions	Slow, inconsistent, depends on operator experience.	Dashboard gives live KPIs, alerts, and recommended action.
Fixed rule-based EMS	Rules are static and do not understand future solar/load conditions.	LSTM forecasts and scenario logic support forward-looking decisions.
Simple solar-following battery logic	Charges/discharges only on current surplus/deficit.	Dispatch also considers tariff, battery safety, future peak windows, and load risk.
SCADA-only monitoring	Shows data but may not optimize.	Combines monitoring with forecasting, metrics, alerts, and control recommendations.

Current/basic solution	Limitation	Project improvement
Offline optimization only	May produce schedules but lacks live operator context.	FastAPI and React dashboard support live refresh and operator interaction.
Black-box AI controller	Operators may not trust unexplained actions.	Dispatch table includes action and reason for every timestep.

Main Differentiator

The strongest differentiator is not simply "we used LSTM" or "we used PPO". Those are known methods. The stronger contribution is the integration:

A safety-bounded, explainable microgrid EMS that combines weather-based forecasting, BESS dispatch, tariff-aware optimization, operator-readable reasoning, scenario testing, MATLAB export, and a live digital-twin simulator.

This makes the project feel closer to a real grid control system than a normal college ML demo.

Current Quantitative Results

From the generated project artifacts:

Metric	Current result
Cleaned NASA rows	52,608
Missing cells after cleaning	0
Duplicate timestamps after cleaning	0
Solar LSTM MAE	2.0748 kW
Solar LSTM RMSE	3.5749 kW
Load LSTM MAE	3.5385 kW
Load LSTM RMSE	4.7597 kW
PPO timesteps	20,000
Seven-day baseline cost	INR 41,165.47
Seven-day optimized cost	INR 36,130.22
Seven-day cost savings	INR 5,035.25
Cost savings percent	12.23 percent
Renewable utilization	93.37 percent
Grid dependency	68.58 percent
Battery safe SoC	True

Real-World Caveat

The current project is a software simulation and control prototype. It is not connected to real smart meters, BMS hardware, inverter telemetry, or certified SCADA. For real deployment, the next step would be hardware-in-the-loop validation, real meter APIs, inverter protocol integration, cyber-security hardening, and operator approval workflows.